A reader's companion to the cutoff-removal theorem for weakly coupled $P(\varphi)_2$

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Read this first. This companion explains a research record and proves nothing new. The record is AI-assisted and has not been refereed; human verification is ongoing. Its main result, Theorem 12.2, is proved conditional on the imported hypothesis U_{GJS} (Assumption A8(vi)), which is not re-derived. Audit item C5, a second human reading of the source page images, is open. Statement numbers refer to `part_i/main.tex`; figures are illustrations, not evidence.

Abstract

A reading guide to a 40-page research record on the spatially cutoff $P(\varphi)_2$ model; it proves nothing new. The record's Theorem 12.2 states that, for $0 \leq \lambda/m_0^2 < \varepsilon_{\mathscr{P}}$, the full net of cutoff ground states ω_g (plateau order, no subnets, no averaging) converges in norm on every local algebra $\mathfrak{A}(O)$. The rate is faster than every inverse power of the distance from O to the plateau edge. The limit is locally normal, translation and (for even $\mathscr{P}$) $\mathbb{Z}_2$ invariant, and a ground state of the Glimm–Jaffe dynamics in the spectral sense. The theorem is proved conditional on U_{GJS}, a cutoff-uniform Glimm–Jaffe–Spencer moment estimate. The record imports it at the scope at which the source states and uses it, without re-deriving it. We explain why the question was open, how a uniform gap and an exact light cone replace compactness, and what is not claimed.

How to read. Fifteen minutes: this page, §1 (what is proved, under what), §3 with Fig. 1 (the architecture) and §8 (what is not claimed). Forty-five minutes: everything except Appendix A, which points into the record for every step not explained here. Every claim carries its locator in the record: statement number and printed page.

1 What is proved, and under what

Theorem 12.2 is proved conditional on one imported estimate, U_{GJS}; the news is that the full net converges, with a rate.

Setting. $\mathcal{H}$ is Q-space over the time-zero free field φ_0 of mass $m_0 > 0$ and covariance $C = (2\mu)^{-1}$, $\mu = (-\partial_x^2 + m_0^2)^{1/2}$. For a bounded open interval O, $\mathfrak{A}(O)$ is the von Neumann algebra generated by the Weyl operators $W(f,h)$, $f, h \in C_c^\infty(O)$ real, and $\mathfrak{A}$ is the norm closure of $\bigcup_O \mathfrak{A}(O)$ [R, §2.1, p. 6]. The interaction sits in the states and in the Glimm–Jaffe dynamics α, a g-independent one-parameter automorphism group of $\mathfrak{A}$ [R, A6(ii), p. 9], not in the local algebras. With $V(h) = \lambda \int h(x) :\mathscr{P}(\varphi_0(x)):_C dx$ (coupling λ, C-Wick

order), where $\mathscr{P}(\xi) = \sum_{n=0}^{2p'} a_n \xi^n$ is the interaction polynomial, set $K(g) = H_0 + V(g)$ (H_0 free), $E(g) = \inf \operatorname{spec} K(g)$ and $H(g) = K(g) - E(g)$; Ω_g is its normalised ground vector, $\omega_g = \langle \Omega_g, \cdot\, \Omega_g \rangle$ the cutoff state, and $\beta_t^g = \operatorname{Ad} e^{itH(g)}$. In [R, Definition 2.1, p. 6], $\mathcal{C}_{\mathrm{E}}$ and $\mathcal{C}_{\mathrm{H}}$ are the measurable $[0,1]$-valued functions of compact support on $\mathbb{R}^2$ and on $\mathbb{R}$, and $\mathcal{G} \subset \mathcal{C}_{\mathrm{H}}$ is the plateau class of $g \in C_c^\infty(\mathbb{R})$ with $0 \le g \le 1$ and $g \equiv 1$ on a nonempty open interval, directed by inclusion of the plateau interiors $\{g = 1\}^\circ$; O_r is the open r-neighbourhood of O, and $d(g, O) = \operatorname{dist}(O, \mathbb{R} \setminus \{g = 1\}^\circ)$. Throughout, $0 \le \lambda/m_0^2 < \varepsilon_{\mathscr{P}} := \lambda_*/M_*^2$, with M_*, λ_* admissible for the Glimm–Jaffe–Spencer (GJS) cluster expansion [7] [R, §2.6, p. 13].

The theorem. Below, τ_a are the space translations and θ is the parity automorphism [R, §2.3, p. 7]; $m_1 > 0$ is the GJS decay constant [R, A8(iv), p. 10]; M is the moment constant of [R, (M_2), p. 16]; and Ψ_{m_1} is the rate function Ψ_γ of [R, (29), p. 28] at the gap parameter $\gamma = m_1$, an explicit function of the distance d stated in §6.

Theorem 12.2 of the record (p. 30)

Assume U_{GJS}, namely Assumption A8(vi), and let $0 \le \lambda/m_0^2 < \varepsilon_{\mathscr{P}}$. Then there is a state ω_∞ on $\mathfrak{A}$ such that:

(1) *(full-net convergence with rate)* for every bounded open interval O, $\lim_{g \in \mathcal{G}} \|(\omega_g - \omega_\infty)|_{\mathfrak{A}(O)}\| = 0$, and $\|(\omega_g - \omega_\infty)|_{\mathfrak{A}(O)}\| \le \Psi_{m_1}(d)$ whenever $g \equiv 1$ on O_d, with Ψ_{m_1} decaying faster than every inverse power;

(2) *(local normality)* $\omega_\infty|_{\mathfrak{A}(O)}$ is normal for every O;

(3) *(ground state)* ω_∞ is a ground state of $(\mathfrak{A}, \alpha)$ in the spectral sense;

(4) *(invariances)* $\omega_\infty \circ \tau_a = \omega_\infty$ for all $a \in \mathbb{R}$; and for even $\mathscr{P}$, $\omega_\infty \circ \theta = \omega_\infty$;

(5) *(spectra and local moment)*

$$\operatorname{spec} H(g) \subseteq \{0\} \cup [m_1, \infty) \quad (g \in \mathcal{C}_{\mathrm{H}}), \qquad \dim \ker H(g) = 1,$$

and $\sup_{g \in \mathcal{C}_{\mathrm{H}}} \|V(h)\Omega_g\| \le M$ for real h with $\|h\|_\infty \le 1$ supported in an interval of length 2.

[...] The theorem is conditional on exactly this list, including U_{GJS}; no other external input is used.

The elided sentences list the imports, printed GJS Theorem 1.1.7, U_{GJS} (A8(vi)) and Assumptions A1–A7, and cite [R, Proposition 3.2, p. 13] for the source's smallness condition. Items (1)–(4) are statements along $\mathcal{G}$; item (5) concerns the cutoff Hamiltonians $H(g)$, $g \in \mathcal{C}_{\mathrm{H}}$.

The import. Write $h_{T,g} = \mathbf{1}_{[-T,T]} \otimes g$ ($T > 0$, $g \in \mathcal{C}_{\mathrm{H}}$) for the slab cutoffs [R, A8(vi), p. 10]. U_{GJS} (A8(vi)) is the cutoff-uniform form of GJS Thm 1.1.8 on the slab family $h_{T,g}$; since 5 September 2026 it is imported at the scope at which the source states it (p. 629) and uses it (pp. 594–597: the affine path and the derivation of Thm 1.1.7 from 1.1.8); it is not a quantifier strengthened beyond the source; it is still a named hypothesis, because it is assembled from four pages and the cluster expansion is not re-derived (Appendix C audits the mechanism and proves two analytic prerequisites only). §4 displays it.

What is new. Convergence, not the topology. Conditional on U_{GJS}, the unaveraged full net $(\omega_g)_{g \in \mathcal{G}}$ has one limit on every local algebra $\mathfrak{A}(O)$, with a rate, and every w^* limit point of the construction in Glimm and Jaffe's paper III [3] (GJ3 below) coincides with ω_∞ on each $\mathfrak{A}(O)$ [R, Corollary 12.3, p. 31]. This makes ω_∞ canonical for the cutoff family.

Uniqueness among the locally normal ground states of $(\mathfrak{A}, \alpha)$, in the spectral sense of [R, Proposition 12.1, p. 29], is not claimed [R, §1.3, p. 4].

CONTEXT, NOT PART OF THE RECORD. In GJ3, the averaged states of [3, (2.4)] lie, on each local algebra, in a norm-compact subset of the local dual, and their limit points are normal [3, Thm. 2.3, p. 211]; in the representation of any w^* limit point, the generator of time translations has spectrum in $[0, \infty)$ [3, Thm. 2.1, p. 211].

Table 1 sorts the statements this companion explains by what their proofs consume, and Fig. 1 shows where U_{GJS} enters and what each item of Theorem 12.2 costs. Status words have one meaning each: proved, partial, conjectured, numerical, refuted; a statement whose proof consumes U_{GJS} is reported as proved, conditional on U_{GJS}.

status	statements of the record (printed page)
proved, conditional on U_{GJS}	Theorems 5.1 (16), 7.8 (21), 11.1 (28), 12.2 (30), 13.2 (31); Lemmas 10.2 (26), 10.3 (27), 10.5 (28); Propositions 10.4 (27), 11.2 (29); Corollary 12.3 (31).
proved, independent of A8 (the GJS package)	Propositions 2.7 (12), 8.1 (22); Lemmas 2.8 (12), 4.4 (16), 9.1 (24), 9.2 (25).
proved as implications under a stated hypothesis	Proposition 8.2 (23) and Lemmas 9.3 (25), 10.1 (26): under the gap hypothesis (G_γ) (22), which for $P(\varphi)_2$ only Theorem 7.8 supplies. Proposition 12.1 (29): for any $(g_n) \subseteq \mathcal{G}$ with $d(g_n, O) \to \infty$ for every bounded interval O and ω_{g_n} converging in norm on every $\mathfrak{A}(O)$.
not claimed; open	The record does not claim uniqueness of ω_∞ among locally normal ground states of $(\mathfrak{A}, \alpha)$ in the spectral sense (Proposition 12.1), a sharp rate, or anything outside the weak-coupling region (Theorem 13.2 is conditional on Hypothesis 13.1); nor a gap for a GNS Hamiltonian of ω_∞, identification with the Euclidean (Wightman) theory of GJS, or Lorentz covariance, which the record does not discuss (§8). *Open:* audit item C5, a transcription check (§9); Level 2, not scheduled (§4); the Part II gate (II.11) (§8.2).

Table 1: Claims ledger, read from the record's proofs. Appendix A classifies every numbered statement of the record.

In the record: §1.3, p. 4; §2.1 and Definition 2.1, p. 6; A6, p. 9; A8(vi) and eq. (U_{GJS}), p. 10; §2.6, p. 13; Proposition 12.1, Theorem 12.2 and Corollary 12.3, pp. 29–31.

2 Why it was open

Compactness gave Glimm and Jaffe a vacuum; it did not give convergence.

Glimm and Jaffe constructed the model in the series I–IV [1, 2, 3, 4]. In paper III, as the record recounts it, the cutoff states are averaged over space translations and a w^*-compactness argument produces a limit point; whether the net (ω_g) itself converges "was not proved there" [R, §1.1, p. 3]. The record quotes the Expositions volume: the $g \to 1$ limit of vacuum expectation values "has not been proved" [6, §4]. It also cites Guerra, Rosen and Simon, who pose Euclidean analogues (local L^p bounds on the Euclidean vacuum density, local convergence of Gibbs states) as Problems 1–2 [8, pp. 125–126]; their note added on p. 128 records Newman's solution of the small-coupling $p = 1$ versions, so "the printed problems must not be read as an unqualified current open-problem claim" [R, §14, p. 32].

Table 2 sets the statements of GJ3 (context) beside those of the record. What the record adds is in the rows *limit* and *selection*, and in the rate of the row *local norm*.

Moments are not enough. Convergence of Schwinger functions at weak coupling follows

property	GJ3 (context)	the record (weak coupling, conditional on U_{GJS})
states	ω_n: averages of ω_{g_n} over space translations, $g_n = g(\cdot/n)$, $g \equiv 1$ on $[-3,3]$ [3, (2.3)–(2.5), p. 210]	ω_g for every $g \in \mathcal{G}$; no averaging [R, Theorem 12.2, p. 30]
limit	a w^* limit point; a subsequence of (ω_n) converges w^* to it [3, Thm. 2.1, p. 211]	the limit of the full net in each $\mathfrak{A}(O)^*$ [R, Theorem 12.2(1), p. 30]
local norm	on each local algebra, (ω_n) lies in a norm-compact subset of the local dual [3, Thm. 2.3, p. 211]	Cauchy in each $\mathfrak{A}(O)^*$; rate $\Psi_{m_1}(d)$ when $g \equiv 1$ on O_d [R, Theorem 11.1, p. 28]
normality	every limit point is normal on each local algebra [3, Thm. 2.3, p. 211]	$\omega_\infty\vert_{\mathfrak{A}(O)}$ is normal [R, Proposition 11.2, p. 29]
energy	in the representation of a w^* limit point, the generator of time translations has spectrum in $[0,\infty)$ [3, Thm. 2.1, p. 211]	ω_∞ is a ground state of $(\mathfrak{A}, \alpha)$ in the spectral sense of [R, Proposition 12.1, p. 29]
selection	the record's reading: compactness "selects a state but proves neither convergence nor canonicity" [R, §1.4, p. 4]	every w^* limit point of GJ3's states equals ω_∞ on each $\mathfrak{A}(O)$ [R, Corollary 12.3, p. 31]

Table 2: GJ3 and the record, statement by statement. The GJ3 column is context; its last cell quotes the record.

from the GJS cluster expansion [R, §1.1, p. 3]: along an affine path of Euclidean cutoffs $h \in \mathcal{C}_{\mathrm{E}}$, GJS bound the change of the moments of products of smeared fields by $O(1)e^{-m_1 d/2}$, with d the distance from the test functions to the change of cutoff, eq. (1.1.11) of [7] as quoted in [R, A8(vii′)(a), p. 11]. In the record's words, such convergence "lives in the wrong topology": "there is no soft passage from moments to norms on local von Neumann algebras" [R, §1.4, p. 4].

Context, not part of the record. The difference $V(g') - V(g)$ has infinite norm, so the machinery for gapped lattice systems of [14, 15] "does not start" [R, §1.4, p. 4].

Later work does not supply the norm statement either, as the record reads it [R, §1.1, p. 3]. Albeverio–Høegh-Krohn–Zegarlinski prove Euclidean Gibbs uniqueness under their stated hypotheses [11], and Bauerschmidt–Dagallier–Weber prove uniqueness of the invariant measure for the Φ^4_2 stochastic-quantization equation above the critical temperature [12]; neither is, by itself, a theorem about local-norm convergence of this Hamiltonian cutoff net. The Hamiltonian renormalisation programme of Rodriguez Zarate–Thiemann is restricted to finite volume [13]. The record identifies no theorem of local-norm convergence "in the references reviewed here"; this is not a priority claim.

In the record: §§1.1 and 1.4, pp. 3–4; A8(vii′)(a), p. 11; Theorem 11.1, Propositions 11.2 and 12.1, Theorem 12.2 and Corollary 12.3, pp. 28–31; §14, p. 32.

3 The architecture

Two cutoff-uniform numbers, the gap m_1 and the moment M, make the net Cauchy.

The record proves Theorem 12.2, conditional on U_{GJS}, in five steps [R, §1.2, p. 3]. Step (i) is the GJS input: the printed clustering estimate CE2 (A8(iv)) and U_{GJS} (A8(vi); §4). The Euclidean layer (the record's §§4–7) derives the uniform local moment (M_2) and the uniform gap (G) by Feynman–Kac–Nelson (FKN) formulas on slabs, steps (ii)–(iii); from these two the Hamiltonian layer (its §§8–11) derives a local-perturbations-perturb-locally

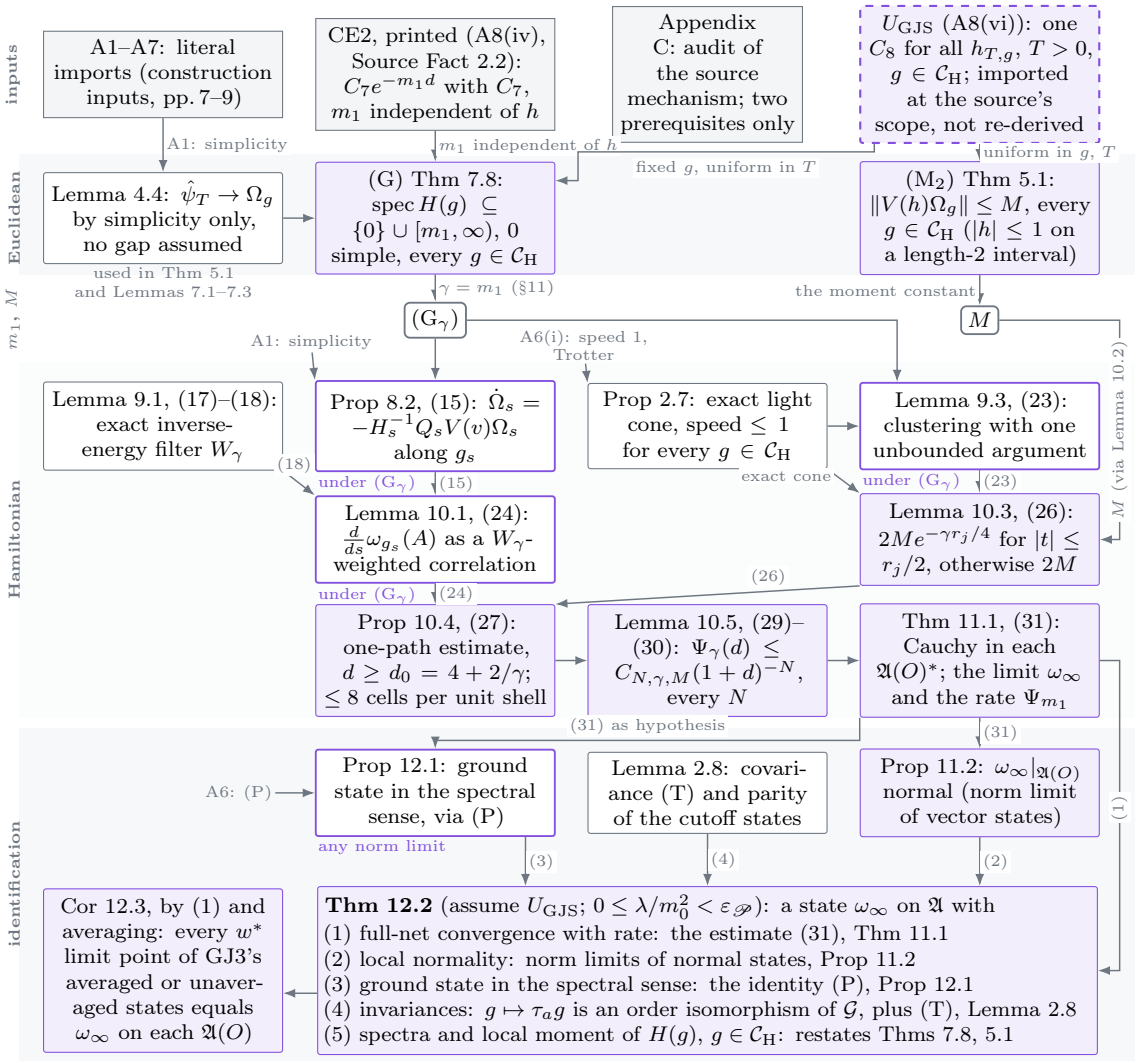


Figure 1: The two layers of the proof and what crosses between them. Boxes name the statements of the record (its Euclidean layer, §§4–7; the interface; its Hamiltonian layer, §§8–11; the identification, §12); arrows say what each proof consumes. Shading records what each proof consumes, read from the record's proofs: violet fill, conditional on U_{GJS}; violet outline, an implication under a hypothesis that for $P(\varphi)_2$ only a violet node supplies; grey, independent of A8. The light-filled inputs of the top row and the two interface nodes carry no class. No arrow from Appendix C closes the dashed box. Drawn from the table of [R, §1.2, p. 3] and the proofs named on each node; nothing is computed. Illustration, not evidence.

(LPPL) estimate that makes the net Cauchy, with a rate, step (iv); its §12 shows that the limit is a ground state of $(\mathfrak{A}, \alpha)$ in the spectral sense (Proposition 12.1), with its invariances, step (v). The convergence argument is Hamiltonian; its two inputs come from Euclidean estimates through slab FKN transfer.

Fig. 1 shades each node by what its proof consumes, read from the record's proofs: violet fill, conditional on U_{GJS}; violet outline, an implication under a hypothesis that for $P(\varphi)_2$ only a violet node supplies; grey, independent of A8. The outlined nodes are Proposition 8.2 and Lemmas 9.3 and 10.1, implications under the hypothesis (G_γ) [R,

§8, p. 22] that $\operatorname{spec} H(g) \subseteq \{0\} \cup [\gamma, \infty)$ and $\dim \ker H(g) = 1$ for every $g \in \mathcal{C}_{\mathrm{H}}$, supplied for $P(\varphi)_2$ only by Theorem 7.8, itself conditional on U_{GJS}; and Proposition 12.1, which holds for any sequence (g_n) in $\mathcal{G}$ with $d(g_n, O) \to \infty$ for every bounded interval O and ω_{g_n} converging in norm on every local algebra $\mathfrak{A}(O)$.

Theorems 5.1 and 7.8, both conditional on U_{GJS}, deliver the two numbers, which Theorem 12.2(5) restates for the cutoff Hamiltonians $H(g)$, $g \in \mathcal{C}_{\mathrm{H}}$ (§1): the moment bound $\|V(h)\Omega_g\| \le M$ for every real h with $\|h\|_\infty \le 1$ supported in an interval of length 2 [R, Theorem 5.1, (M_2), p. 16], and the gap $\operatorname{spec} H(g) \subseteq \{0\} \cup [m_1, \infty)$ [R, Theorem 7.8, (G), p. 21].

The record's proofs show where each uniformity comes from. The g-independence of m_1 is CE2's printed h-independence [R, Source Fact 2.2, p. 10], applied across a horizontal strip [R, Lemma 7.4, p. 19]. In the record's §7, U_{GJS} is used at one fixed g, uniformly in T [R, §7, opening sentence, p. 18], by Lemmas 7.1–7.3, which are thus conditional on U_{GJS}. Across g, U_{GJS} makes M independent of g; Lemma 10.2, conditional on U_{GJS} through Theorem 5.1, uses this at every cutoff g_s of an affine path [R, Lemma 10.2, p. 26].

The record's §§8–10 work with an abstract gap γ under (G_γ), and its §§8–11 use the GJS package only through m_1 and M: Lemma 10.2 calls Theorem 5.1, and γ is instantiated as m_1 in [R, §11, p. 28]. Their output, the one-path estimate [R, Proposition 10.4, eq. (27), p. 27], conditional on U_{GJS} through M, bounds $\|(\omega_g - \omega_{g'})|_{\mathfrak{A}(O)}\|$ whenever $g \equiv g' \equiv 1$ on O_d, $d \ge d_0 := 4 + 2/\gamma$. Plateaus are cofinal and local duals are complete, so the net converges in norm in each $\mathfrak{A}(O)^*$, at a rate Ψ_{m_1} faster than every inverse power (§6):

Checkpoint: Theorem 11.1, conditional on U_{GJS}

$$\|(\omega_g - \omega_\infty)|_{\mathfrak{A}(O)}\| \le \Psi_{m_1}(d) \qquad \text{whenever } g \equiv 1 \text{ on } O_d.$$

[R, Theorem 11.1, eq. (31), p. 28]

In the record: §1.2, p. 3; Source Fact 2.2, p. 10; Theorem 5.1, Lemmas 7.1–7.4 and Theorem 7.8, pp. 16–21; (G_γ), p. 22; Lemma 10.2, Proposition 10.4 and Theorem 11.1, pp. 26–28.

4 The one import that is not a quotation

U_{GJS} is the source's own quantifier, assembled from four pages and not re-derived; that is why the theorem is conditional.

Of the record's eight imports [R, §2.4, p. 7], A1–A7 quote printed construction theorems (table below). A8 packages the cluster-expansion input of Glimm–Jaffe–Spencer [7] for the interacting Euclidean measures dq_h, $h \in \mathcal{C}_{\mathrm{E}}$, of A3 and for observables $Q(x^0)$ at sharp Euclidean time x^0; its items include the printed norms $\|\cdot\|_{\mathrm{GJS}}$, the clustering estimate CE2 (Source Fact 2.2), the moment bound CE1 (Source Fact 2.3), the dilation, and one item not read off a single theorem:

Assumption A8(vi) of the record (p. 10)

For

$$\mathcal{C}_{\mathrm{E}}^{\mathrm{slab}} := \{h_{T,g} = \mathbf{1}_{[-T,T]} \otimes g : T > 0,\ g \in \mathcal{C}_{\mathrm{H}}\} \subset \mathcal{C}_{\mathrm{E}},$$

one and the same constant C_8 satisfies

$$\left| \int Q(x^0)\, dq_{h_{T,g}} \right| \le C_8 \|Q(x^0)\|_{\mathrm{GJS}} \quad (T > 0,\ g \in \mathcal{C}_{\mathrm{H}}). \qquad [\mathrm{R}, (U_{\mathrm{GJS}}), \text{p. } 10]$$

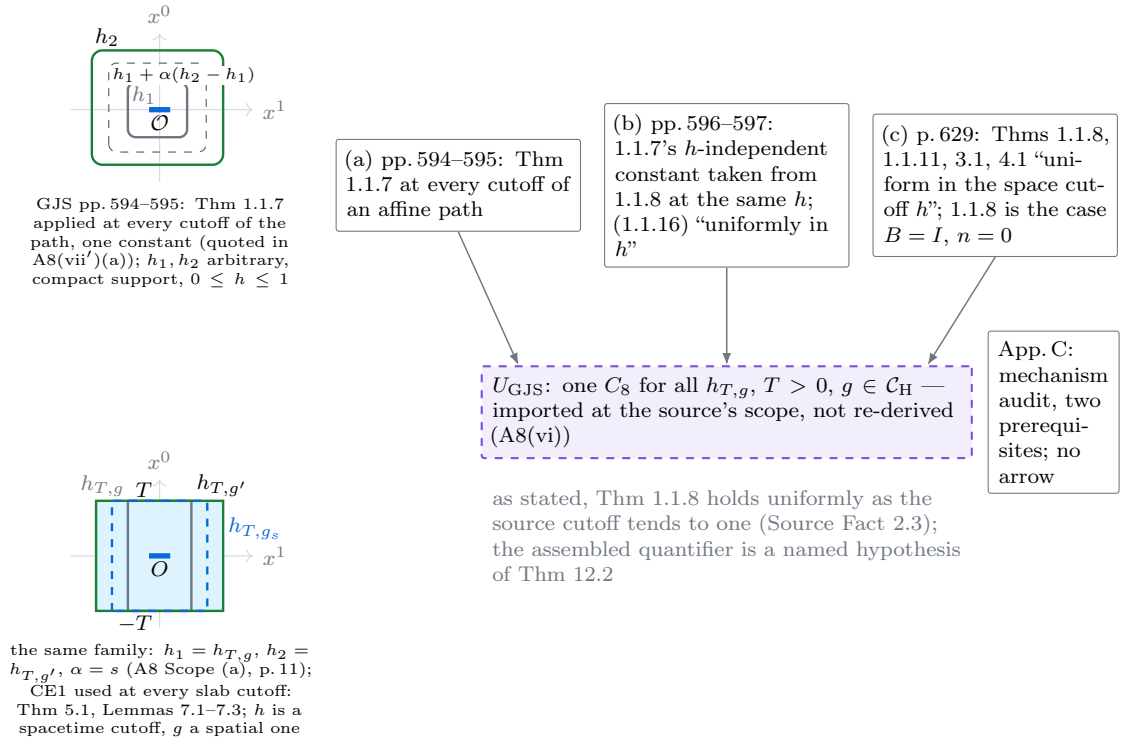


Figure 2: Two affine paths, one quantifier. Left: the source's path between two cutoffs h_1, h_2 of its class (top) and the record's slab path h_{T,g_s} (bottom); h is a spacetime cutoff, g a spatial one. Right: the three steps of the record's reading, [R, A8, Scope paragraph, pp. 11–12]. The source states this quantifier (p. 629) and uses it (pp. 594–597); the record imports it at that scope and does not re-derive it, so U_{GJS} remains a named hypothesis of Theorem 12.2. Audit item C5, a second human reading of these pages, is open. Nothing is computed, and no page image is reproduced. Illustration, not evidence.

U_{GJS} (A8(vi)) is the cutoff-uniform form of GJS Thm 1.1.8 on the slab family $h_{T,g}$; since 5 September 2026 it is imported at the scope at which the source states it (p. 629) and uses it (pp. 594–597: the affine path and the derivation of Thm 1.1.7 from 1.1.8); it is not a quantifier strengthened beyond the source; it is still a named hypothesis, because it is assembled from four pages and the cluster expansion is not re-derived (Appendix C audits the mechanism and proves two analytic prerequisites only).

Why it is not literal. As printed, GJS Theorem 1.1.8 is "stated uniformly as the source cutoff tends to one" [R, Source Fact 2.3, p. 10]. The record needs it on another family: its limits $T \to \infty$ are taken at fixed g [R, §4, p. 15], the bound must hold along every affine cutoff $g_s = (1 - s)g + sg'$ [R, A8(vi), p. 10], and the slab cutoffs $h_{T,g}$, with g of compact support, do not tend to one. Hence Theorem 12.2 is proved, conditional on U_{GJS} [R, Theorem 12.2, p. 30].

The record's reading. The Scope paragraph of A8 argues from the passages that A8(vii′) quotes [R, A8(vii′), pp. 10–11], in three steps (Fig. 2):

(a) the record reads GJS pp. 594–595 as proving Theorem 1.1.1 along the affine path $h_1 + \alpha(h_2 - h_1)$, $\alpha \in [0, 1]$, for arbitrary cutoffs h_1, h_2 of the source's class (p. 589: compact support, $0 \le h \le 1$), with Theorem 1.1.7 applied at every cutoff of that path with one h-independent constant; the record's path $h_{T,g_s} = (1 - s)h_{T,g} + s\, h_{T,g'}$ is this family with $\alpha = s$ [R, A8, Scope (a), p. 11];

(b) it reads GJS pp. 596–597 as taking the h-independent constant of Theorem 1.1.7 from Theorem 1.1.8 and (1.1.16) at the same h; since Lemma 7.4 uses Theorem 1.1.7 as printed at $h = h_{T,g}$, it concludes that consistency requires the constant of Theorem 1.1.8 to be uniform on the same class, which is (U_{GJS}) [R, A8, Scope (b), pp. 11–12];

(c) it reads GJS p. 629, as quoted in [R, Source Fact 2.4, p. 10], as stating the uniformity of Theorems 1.1.8 and 1.1.11 in the space cutoff h, with Theorem 1.1.8 the case $B = I$, $n = 0$ [R, A8, Scope (c), p. 12].

Where it is consumed. Proposition C.7 of the record, conditional on U_{GJS} like Theorem 5.1 and Lemmas 7.1–7.3, lists every application [R, Proposition C.7, p. 38]: CE1, through (U_{GJS}), in Theorem 5.1 and Lemmas 7.1–7.3, each time at a slab cutoff; CE2 in Lemma 7.4 only, as printed, with constants independent of h.

What Appendix C of the record does. It audits five source mechanisms, and its Proposition C.1 is "deliberately limited to the audited mechanisms" [R, Proposition C.1, p. 36]; it proves h-uniform normalisation and stability [R, Lemma C.2, p. 37], [R, Theorem C.5, p. 38]. It re-derives neither the expansion nor U_{GJS} [R, Appendix C, p. 35], and Theorem 12.2 does not consume it.

C5 and the routes out. Audit item C5, a second human reading of the GJS page images against the transcriptions, is open [R, Appendix B, "External residue", p. 35]: its nine items have blank verdicts (`part_i/C5_reading_record.md`). It is a transcription check; closing it leaves U_{GJS} a hypothesis (`docs/status.md`). The only proportionate route, which `strategy.md` calls Level 2, is recorded in `docs/imports.md`: an h-trace of the printed expansion on GJS pp. 598–629, completing Prop C.1 (not scheduled). The Hamiltonian bypass for (M_2) is recorded as dead in `strategy.md`, since g-uniform local moments were open in the 1970–71 literature; `docs/imports.md` calls this "a verdict on the printed record, not a no-go theorem".

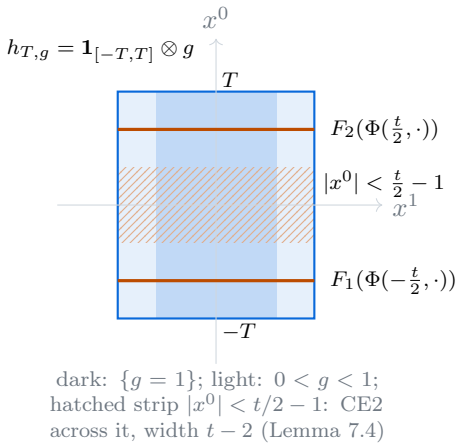
The seven literal imports [R, §2.4, p. 7]:

	import: what it supplies	source pinpoint
A1	*cutoff construction:* the ground state Ω_g of $H(g)$; the closure of $K(g)$ "has $E(g) = \inf \operatorname{spec} K(g)$ as a *simple* eigenvalue"	[7, p. 588]; [2, Thms. 2.2.1, 2.3.1]; [5]; [9, Thm. 5.3]
A2	*Feynman–Kac–Nelson transfer:* the slab objects $h_{T,g}$, ψ_s, c_g and the Euclidean $\leftrightarrow$ Hamiltonian passage	[10, Thm. III.6, Cor. V.14, Thm. V.15]; [8, Thms. II.14, II.16]
A3	*finite-volume stability:* the Euclidean interaction $\mathcal{U}_h$ in every L^p, $e^{-\varrho \mathcal{U}_h}$ integrable ($\varrho > 0$), so the measures dq_h are well defined	[10, Thm. V.7]
A4	*free hypercontractivity:* the L^p smoothing used in the pairing bound (the record's §8) and in Appendix C	[10, Thms. I.17, III.17]
A5	*domain smoothing; common core:* $\Omega_g \in D(V(h))$, and a common core for all $K(g_s)$ along an affine path	(i) [6, Ch. 7, Thm. 7.4]; (ii) [10, Thm. V.12], [9, Thm. 5.3], [2]
A6	*free propagation; patching; the dynamics α:* (i) speed one, the Trotter formula; (ii) (P): $\alpha_t(\mathfrak{A}(O)) \subseteq \mathfrak{A}(O_{\lvert t\rvert})$, $\alpha_t(A) = \beta_t^g(A)$ if $g \equiv 1$ on $O_{\lvert t\rvert}$ ($A \in \mathfrak{A}(O)$); the cutoff cone is Proposition 2.7	[1, §III]; [2, §§3.5–3.6]; [6, §4.1 and Ch. 4]
A7	*covariance data:* translation covariance and parity	[2, p. 398]

In the record: A1–A8 with Source Facts 2.2–2.4, eq. (U_{GJS}), item (vii′) and the Scope paragraph, pp. 7–12; Appendix B, p. 35; Appendix C with Proposition C.1, Lemma C.2, Theorem C.5 and Proposition C.7, pp. 35–38.

5 Euclidean layer: the moment and the gap

The gap is an output: simplicity of the ground state is the only spectral input.



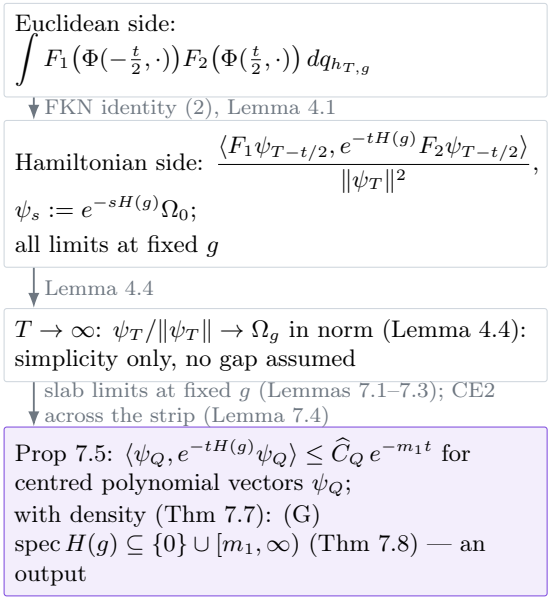


Figure 3: Slab transfer, and the gap as an output. Left: the slab cutoff $h_{T,g}$, the two sharp times of [R, Lemma 4.1, eq. (2), p. 15], and the strip across which CE2 is applied [R, Lemma 7.4, p. 19]. Right: the Hamiltonian side of (2), the limit $T \to \infty$ [R, Lemma 4.4, p. 16], and the output, [R, Proposition 7.5, p. 20] and [R, Theorem 7.8, p. 21], conditional on U_{GJS} through Lemmas 7.1–7.3. Every limit $T \to \infty$ is taken at fixed g, and no gap is assumed: the gap of Theorem 7.8 is the output of this picture. Working coordinates; lengths are not physical. Nothing is computed. Illustration, not evidence.

This layer produces, one cutoff $g \in \mathcal{C}_{\mathrm{H}}$ at a time, the moment M and the gap m_1 that §6 consumes; every limit $T \to \infty$ is taken at fixed g, and no gap is assumed [R, §4, p. 15]. Theorem 5.1, Lemmas 7.1–7.3, Proposition 7.5, Lemma 7.6 and Theorems 7.7–7.8 are conditional on U_{GJS} (A8(vi); §4); Fig. 3 draws the slab.

I. Slab transfer. With $\Omega_0 \equiv 1$ the free vacuum, $\psi_s := e^{-sH(g)}\Omega_0$ and $c_g := \langle\Omega_g, \Omega_0\rangle > 0$, the FKN import A2 turns sharp-time moments of the Euclidean field Φ under the normalised slab measure $dq_{h_{T,g}}$ into matrix elements of $e^{-tH(g)}$ [R, Lemma 4.1, p. 15]. As $T \to \infty$, $\hat{\psi}_T := \psi_T/\|\psi_T\|$ converges to Ω_g: ψ_s is $c_g\Omega_g$ plus $e^{-sH(g)}$ applied to the component of Ω_0 orthogonal to Ω_g, whose spectral measure has no atom at 0 by simplicity (A1).

Checkpoint I: transfer at fixed g

For $T > 0$, $t \in [0, 2T]$ and bounded nonnegative measurable F_1, F_2 of the time-zero field,

$$\int F_1\big(\Phi(-\tfrac{t}{2},\cdot)\big)F_2\big(\Phi(\tfrac{t}{2},\cdot)\big)\,dq_{h_{T,g}} = \frac{\langle F_1\psi_{T-t/2}, e^{-tH(g)}F_2\psi_{T-t/2}\rangle}{\|\psi_T\|^2}, \qquad \text{[R, (2), p. 15]}$$

and, as $s \to \infty$, "$\hat{\psi}_s \to \Omega_g$ in norm, $\|\psi_s\| \downarrow c_g$"; "*No isolation of the ground-state eigenvalue is used.*" [R, Lemma 4.4, p. 16]

II. The moment. For h as in (M_2) below, Steps 1–4 of the proof of Theorem 5.1 treat $Q_V = V(h)^2$ as one observable of the GJS class, which admits products of individually Wick-ordered powers at equal times [R, A8(ii), p. 9]: the support of h meets at most three

unit cells, hence at most nine ordered cell pairs, and the printed norm, with constants K_1 and $K_2 \geq 1$ [R, A8(iii), p. 10], gives $\|Q_V\|_{\mathrm{GJS}} \leq M^2/C_8$ [R, Theorem 5.1, eq. (5), p. 17]. Eq. (2) at $t = 0$ with $F_1 = F_2 = |V(h)|$ and (U_{GJS}) at $h_{T,g}$ give $\|V(h)\hat{\psi}_T\| \leq M$ for every T [R, eq. (6), p. 17]; the graph of $V(h)$ is weakly closed and $\hat{\psi}_T \to \Omega_g$, so it passes to Ω_g:

Checkpoint II: the uniform local moment

Let $0 \leq \lambda/m_0^2 < \varepsilon_{\mathscr{P}}$. For every $g \in \mathcal{C}_{\mathrm{H}}$ and every real measurable h with $\|h\|_\infty \leq 1$ and $\operatorname{supp} h$ contained in an interval of length 2,

$$\|V(h)\Omega_g\| \leq M, \qquad M^2 := 9\, C_8\, \mathfrak{K}(p')\, \lambda^2 \Big(\sum_{n=0}^{2p'} |a_n|\Big)^2, \qquad \mathfrak{K}(p') := K_1 K_2^{8p'} (4p')!.$$

[R, (M_2), p. 16]

III. Decay at fixed g. Lemmas 7.1–7.3 pass the moments of a cylinder monomial Q of sharp-time fields from the slab measures to Ω_g as $T \to \infty$, with (U_{GJS}) bounding the square moments at one g, uniformly in T [R, Lemmas 7.1–7.3, pp. 18–19]. Separately, CE2, with constants C_7 and m_1 [R, Source Fact 2.2, p. 10], applied across the strip $\{|x^0| < t/2 - 1\}$ of width $t-2$, bounds the connected correlation of $Q(-t/2)$ and $Q(t/2)$ by $C_7 e^{2m_1} e^{-m_1 t} \|Q\|_{\mathrm{GJS}}^2$ for $t > 2$ and every cutoff [R, Lemma 7.4, p. 19]. Together, at fixed g:

Checkpoint III: semigroup decay of centered observables

Let $\psi_Q := Q\Omega_g - \langle \Omega_g, Q\Omega_g \rangle \Omega_g$. Then

$$\langle \psi_Q, e^{-tH(g)} \psi_Q \rangle \leq \widehat{C}_Q\, e^{-m_1 t} \quad (t \geq 0), \qquad \widehat{C}_Q := \max\big\{ C_7 e^{2m_1} \|Q\|_{\mathrm{GJS}}^2,\ e^{3m_1} \|\psi_Q\|^2 \big\}.$$

[R, Proposition 7.5, p. 20]

IV. The gap. Decay on centred vectors controls the spectrum only if, with Ω_g, they span a dense subspace. They do [R, Theorem 7.7, p. 21], by exponential moments of the sharp-time fields in the vector state of Ω_g [R, Lemma 7.6, p. 20], which use powers of every total degree as the source's class admits [R, A8(ii), p. 9], strip-analytic characteristic functionals, Fourier uniqueness and martingale convergence. For every $0 \leq \beta < m_1$, Checkpoint III forces $\mathbf{1}_{[0,\beta]}(H(g))$ to annihilate every ψ_Q, so by density it is the projection onto $\mathbb{C}\Omega_g$; that is, $H(g)$ has no spectrum in $(0, m_1)$ [R, Theorem 7.8, p. 21]:

Checkpoint IV: the uniform gap

Let $0 \leq \lambda/m_0^2 < \varepsilon_{\mathscr{P}}$. For every $g \in \mathcal{C}_{\mathrm{H}}$,

$$\operatorname{spec} H(g) \subseteq \{0\} \cup [m_1, \infty), \qquad \text{[R, (G), p. 21]}$$

and 0 is a simple eigenvalue.

No gap entered Pieces I–III; (G) is their output. The slab limits use weak compactness in $\mathcal{H}$, and each limit is identified by a closed graph, so nothing is selected.
In the record: A8(ii)–(iii) and Source Fact 2.2, pp. 9–10; Lemmas 4.1 and 4.4, Theorem 5.1 and eqs. (2), (5), (6), (M_2), pp. 15–17; Lemmas 7.1–7.4 and 7.6, Proposition 7.5, Theorems 7.7–7.8 and (G), pp. 18–21.

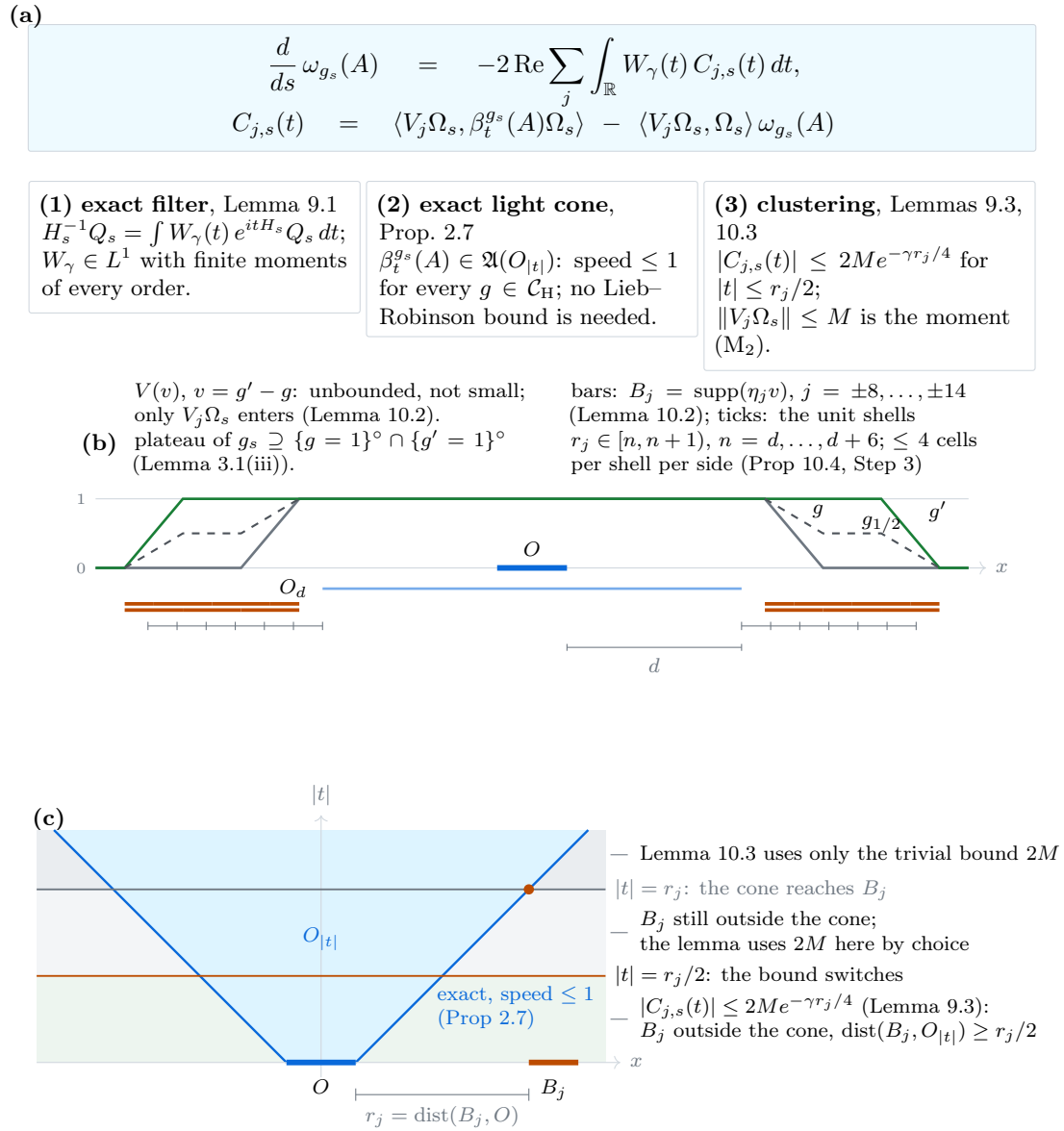


Figure 4: The derivative of the state along the path. (a) The three moves that control it, [R, Lemma 10.1, eq. (24), p. 26] and [R, Proposition 10.4, Step 2, p. 27]. (b) The path g_s and the cells B_j of [R, Lemma 10.2, p. 26], drawn only where $v = g' - g \neq 0$, with the unit shells of [R, Proposition 10.4, Step 3, p. 27]. (c) The two-regime bound of [R, Lemma 10.3, p. 27], drawn to scale for one cell, with the exact light cone of [R, Proposition 2.7, p. 12]. The switch at $|t| = r_j/2$ is a switch of the bound, not of the correlation; the cone reaches B_j only at $|t| = r_j$. No automorphism or spectral flow is constructed: the derivative of the state is bounded and integrated along the path. Nothing is computed. Illustration, not evidence.

6 Hamiltonian layer: local perturbations perturb locally

An exact filter turns the gap into decay in time; an exact light cone turns time into distance.

For $g, g' \in \mathcal{G}$, every $\gamma > 0$ satisfying (G_γ) [R, §8, p. 22], and M the moment constant of (M_2), the record's §§8–10 prove the following contract, with Ψ_γ the rate function of [R, (29), p. 28], assembled at the end of this section; through M it is conditional on U_{GJS}.

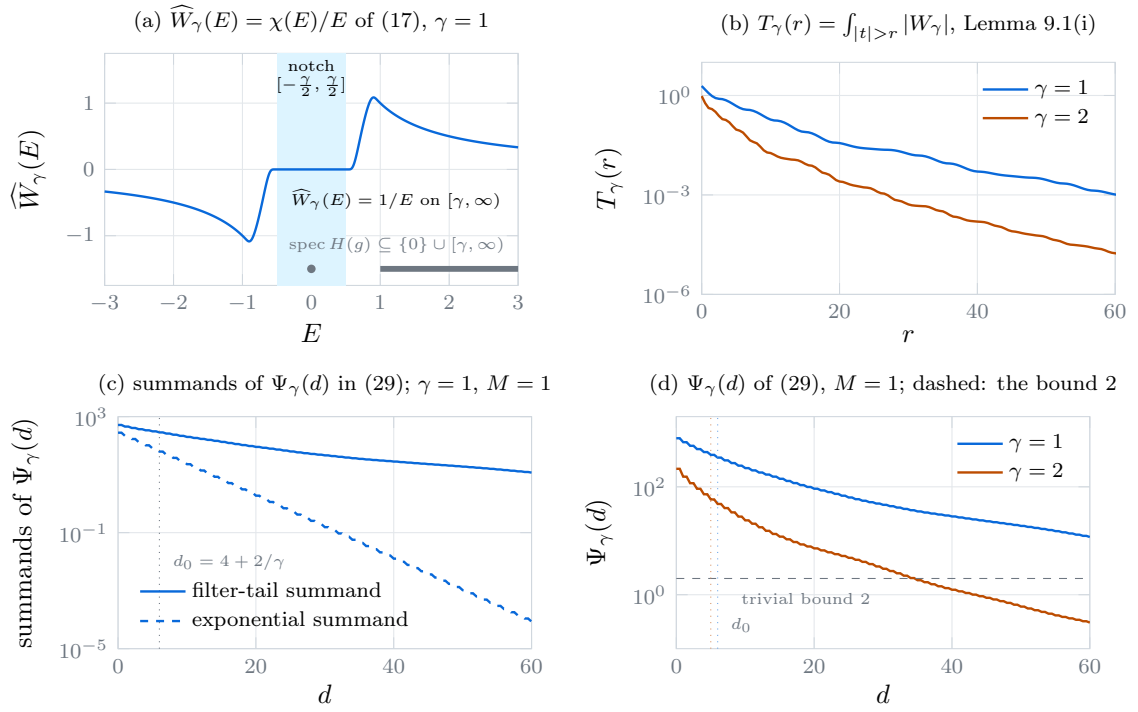


Figure 5: Filter, tail and rate. (a) $\widehat{W}_\gamma(E) = \chi(E)/E$ of [R, Lemma 9.1, eq. (17), p. 24] with its notch; (b) the tail $T_\gamma(r)$ of [R, Lemma 9.1(i), p. 24]; (c) the two summands of $\Psi_\gamma(d)$, [R, Lemma 10.5, eq. (29), p. 28]; (d) $\Psi_\gamma(d)$ with the trivial bound and $d_0 = 4 + 2/\gamma$. Computed by `docs/figures/make_figures.py` for one admissible χ, $\gamma \in \{1, 2\}$ and $M = 1$, on a time grid up to $t = 160$, with values of $|W_\gamma|$ below 10^{-9} set to zero. Lemma 9.1(i) proves only that $W_\gamma \in L^1$ with finite moments of every order, hence $T_\gamma(r) \le C_{q,\gamma}(1 + r)^{-q}$; the curves are numerical and claim no decay law. Ψ_γ is an upper bound on $\|(\omega_g - \omega_{g'})|_{\mathfrak{A}(O)}\|$, never a measured decay of any state; $M = 1$ is a placeholder, not the record's M, which depends on unquantified GJS constants. At $\gamma = 1$, $\Psi_\gamma(d)$ exceeds the trivial bound 2 at every d shown; at $\gamma = 2$ it falls below 2 from $d = 35$ on: the content is the decay in d, not the size at any d. Both summands decay faster than every inverse power; the exponential summand decays faster, so the filter-tail summand sets the rate. Illustration, not evidence.

The contract, Lemma 10.5

$$\|(\omega_g - \omega_{g'})|_{\mathfrak{A}(O)}\| \le \Psi_\gamma(d) \quad \text{whenever } g \equiv g' \equiv 1 \text{ on } O_d,\ d \ge 0, \qquad \text{[R, Lemma 10.5, p. 28]}$$

and for every $N \ge 1$ there is $C_{N,\gamma,M} < \infty$ with

$$\Psi_\gamma(d) \le C_{N,\gamma,M}\,(1 + d)^{-N} \qquad (d \ge 0). \qquad \text{[R, (30), p. 28]}$$

Fig. 4 draws the three moves that prove it.

V. The derivative along a path. Put $g_s := (1 - s)g + sg'$ and $v := g' - g$, $s \in [0, 1]$. If $g \equiv g' \equiv 1$ on O_d, every g_s lies in $\mathcal{G}$ and equals 1 on O_d [R, Lemma 3.1(iii), p. 13]; s is a path parameter, not a time. Proposition 8.2, an implication under (G_γ), differentiates the ground vector $\Omega_s := \Omega_{g_s}$ of $H_s := H(g_s)$ in the parallel gauge: $\dot{\Omega}_s = -H_s^{-1} Q_s V(v) \Omega_s$, with $Q_s := 1 - |\Omega_s\rangle\langle\Omega_s|$ and $\|H_s^{-1} Q_s\| \le \gamma^{-1}$ [R, Proposition 8.2, eq. (15), p. 23]. The exact inverse-energy filter of [R, Lemma 9.1, p. 24] trades this reduced resolvent for real-time evolution: for an even $\chi \in C^\infty(\mathbb{R}; [0, 1])$ with $\chi = 0$ on $[-\gamma/2, \gamma/2]$ and $\chi = 1$ outside

$(-\gamma,\gamma)$, let W_γ be the inverse Fourier transform of $\widehat{W}_\gamma(E) := \chi(E)/E$ $(E \neq 0)$, $\widehat{W}_\gamma(0) := 0$ [R, eq. (17), p. 24]. If H is self-adjoint with $\operatorname{spec} H \subseteq \{0\} \cup [\gamma,\infty)$, then

$$H^{-1}(1 - P_{\ker H}) = \int_{\mathbb{R}} W_\gamma(t)\, e^{itH}(1 - P_{\ker H})\, dt, \qquad \text{[R, Lemma 9.1(ii), eq. (18), p. 24]}$$

an integral identity applied to vectors. Lemma 10.1, an implication under (G_γ), applies it with $H = H_s$ to the single vector $Q_s A\Omega_s$; for self-adjoint $A \in B(\mathcal{H})$ and $0 < s < 1$:

Checkpoint: the state derivative, Lemma 10.1

$$\frac{d}{ds}\,\omega_{g_s}(A) = -2\,\mathrm{Re}\int_{\mathbb{R}} W_\gamma(t)\Big[\langle V(v)\Omega_s, \beta_t^{g_s}(A)\Omega_s\rangle - \langle V(v)\Omega_s, \Omega_s\rangle\,\omega_{g_s}(A)\Big]\,dt.$$

[R, (24), p. 26]

VI. From time to distance. Every cutoff dynamics has an exact light cone: $\beta_t^g(\mathfrak{A}(O)) \subseteq \mathfrak{A}(O_{|t|})$ for all $g \in \mathcal{C}_{\mathrm{H}}$ and $t \in \mathbb{R}$, with speed ≤ 1 independent of g, from A6(i) [R, Proposition 2.7, p. 12]. The perturbation $V(v)$ is unbounded, and (M_2) controls only pieces supported in intervals of length 2, so it is cut into cells $V_j := V(\eta_j v)$, $B_j := \operatorname{supp}(\eta_j v)$, by a smooth partition of unity $\sum_j \eta_j = 1$ with $0 \leq \eta_j \leq 1$ and $\operatorname{supp}\eta_j \subset (j-1, j+1)$: then $V(v)\xi = \sum_j V_j\xi$ on common domains and $\|V_j\Omega_s\| \leq M$ for $0 \leq s \leq 1$ [R, Lemma 10.2, eq. (25), p. 26], a lemma conditional on U_{GJS} through Theorem 5.1. Only $V_j\Omega_s$ enters, never $\|V_j\|$. Lemma 9.3, an implication under (G_γ), allows one unbounded argument X: its proof bounds at $t = 0$ two functions of t with spectral supports in $[\gamma,\infty)$ and $(-\infty,-\gamma]$, which the light cone makes agree for $|t| < R$, by the spectral separation [R, Lemma 9.2, eq. (19), p. 25]; the bound needs $\|X\Omega\|$, not $\|X\|$:

Checkpoint: clustering with one unbounded argument, Lemma 9.3

Let $g \in \mathcal{C}_{\mathrm{H}}$, write $H = H(g)$, $\Omega = \Omega_g$, $\beta_t = \beta_t^g$, and assume (G_γ). Let $B \subset \mathbb{R}$ be compact, let $X = X^*$ be affiliated with $\mathfrak{A}(U)$ for every bounded open $U \supseteq B$, with $\Omega \in D(X)$, and let $A \in \mathfrak{A}(O)$ for a bounded interval O. If $R := \operatorname{dist}(B, O) > 0$ satisfies $\gamma R \geq 1$, then

$$\big|\langle X\Omega, A\Omega\rangle - \langle X\Omega,\Omega\rangle\langle\Omega, A\Omega\rangle\big| \leq 2\,\|X\Omega\|\,\|A\|\,e^{-\gamma R/2}. \qquad \text{[R, (23), p. 25]}$$

VII. Two regimes, then shells. Fix $A = A^* \in \mathfrak{A}(O)$ with $\|A\| \leq 1$. For a nonzero cell put $r_j := \operatorname{dist}(B_j, O)$ and $C_{j,s}(t) := \langle V_j\Omega_s, \beta_t^{g_s}(A)\Omega_s\rangle - \langle V_j\Omega_s, \Omega_s\rangle\,\omega_{g_s}(A)$, the cell's share of the bracket in (24). If $\gamma r_j/2 \geq 1$, Lemma 10.3, conditional on U_{GJS} through M, applies (23) with $X = V_j$ and gives

$$|C_{j,s}(t)| \leq \begin{cases} 2Me^{-\gamma r_j/4}, & |t| \leq r_j/2, \\ 2M, & t \in \mathbb{R}. \end{cases} \qquad \text{[R, Lemma 10.3, eq. (26), p. 27]}$$

For $|t| \leq r_j/2$ the cell stays at distance $\geq r_j/2$ from $O_{|t|}$; beyond that, Lemma 10.3 uses only the trivial bound $2M$ (by choice; the cone reaches B_j at $|t| = r_j$). If $d \geq d_0 := 4 + 2/\gamma$ and $g \equiv g' \equiv 1$ on O_d, every cell has $r_j \geq d$, so $\gamma r_j/2 \geq 2\gamma + 1$ and Lemma 10.3 applies to every cell. Splitting each time integral of (24) at $|t| = r_j/2$ weights the inner bound by $\|W_\gamma\|_1$ and the outer one by the filter tail $T_\gamma(r_j/2)$, where $T_\gamma(r) := \int_{|t|>r} |W_\gamma(t)|\,dt$. Summing the cells shell by shell, at most eight with r_j in any unit shell $[n, n+1)$, four on each side (Fig. 4(b)), and integrating over s gives Proposition 10.4, conditional on U_{GJS} through M (Steps 3–4 of its proof):

Checkpoint: the one-path estimate, Proposition 10.4

For $d \geq d_0$ and $g \equiv g' \equiv 1$ on O_d,

$$\|(\omega_g - \omega_{g'})|_{\mathfrak{A}(O)}\| \leq 32M \sum_{n \geq \lfloor d \rfloor} \Big(\|W_\gamma\|_1 e^{-\gamma n/4} + T_\gamma(n/2) \Big). \qquad \text{[R, (27), p. 27]}$$

The rate. Lemma 10.5 names the right side of (27) $\Psi_\gamma(d)$ for $d \geq d_0$, and sets $\Psi_\gamma(d) := \max\{2, \text{the same expression}\}$ for $0 \leq d < d_0$ [R, eq. (29), p. 28]. The exponential summand is at most a constant times $e^{-\gamma d/4}$. Lemma 9.1(i) is credited only with $W_\gamma \in L^1$ and finite moments of every order, hence $T_\gamma(r) \leq C_{q,\gamma}(1+r)^{-q}$; this controls the filter-tail summand. Together they give (30), the second display of the contract above. These constants, and Ψ_γ itself, depend on the fixed choice of χ. Both summands decay faster than every inverse power; the exponential summand decays faster, so the filter-tail summand sets the rate. W_γ cannot have exponential tails, because $\widehat{W_\gamma}$ vanishes on $[-\gamma/2, \gamma/2]$ by construction and a transform analytic in a strip cannot vanish on an interval without vanishing identically [R, §1.4, p. 5]. The record states that "Ψ_{m_1} is super-polynomial but not exponential, and this is intrinsic to the filter method used" [R, §1.3(b), p. 4]. Ψ_γ is an upper bound on a norm difference, never a measured decay of any state; Fig. 5 draws its ingredients for one admissible χ.

Context, not part of the record. The machinery of [14, 15] "does not start" here, because $V(g') - V(g)$ has infinite norm (§2; [R, §1.4, p. 4]). No automorphism or spectral flow is constructed: the derivative of the state is bounded and integrated along the path. And no Lieb–Robinson bound is needed: the light cone of Proposition 2.7 is exact.

In the record: §§1.3(b) and 1.4, pp. 4–5; Proposition 2.7, p. 12; Lemma 3.1(iii), p. 13; (G_γ), Proposition 8.2, Lemmas 9.1–9.3 and 10.1–10.3, Proposition 10.4 and Lemma 10.5, with eqs. (15), (17)–(19), (23)–(27) and (29)–(30), pp. 22–28.

7 From Cauchy to the theorem

The ground-state property costs an identity, not an estimate.

Everything in this section is conditional on U_{GJS} through Theorem 11.1, except Proposition 12.1 and Lemma 2.8, which do not use A8. Theorem 11.1 turns the one-path estimate of §6 into convergence [R, Theorem 11.1, p. 28]. Step 1: if $\Psi_{m_1}(d) < \varepsilon$, all $g, g' \in \mathcal{G}$ whose plateau interiors contain O_d satisfy $\|(\omega_g - \omega_{g'})|_{\mathfrak{A}(O)}\| \leq \Psi_{m_1}(d) < \varepsilon$ [R, (32), p. 28], so the net is Cauchy in each $\mathfrak{A}(O)^*$. Step 2: completeness of each $\mathfrak{A}(O)^*$ gives limits, compatible under restriction, that define a state ω_∞ on $\mathfrak{A}$. Step 3: the cutoffs equal to 1 on O_d are cofinal in $\mathcal{G}$, so the bound passes to the limit; this is the checkpoint (31) displayed at the end of §3 [R, (31), p. 28].

Local normality comes with the topology: Proposition 11.2 obtains it from the normality of the vector states ω_g on the von Neumann algebra $\mathfrak{A}(O)$ [R, Proposition 11.2, p. 29].

The dynamics α is imported: A6(ii) supplies a one-parameter automorphism group of $\mathfrak{A}$, independent of g, with $\alpha_t(\mathfrak{A}(O)) \subseteq \mathfrak{A}(O_{|t|})$ and the patching property (P): $\alpha_t(A) = \beta_t^g(A)$ for $A \in \mathfrak{A}(O)$ whenever $g \equiv 1$ on $O_{|t|}$ [R, A6, p. 9]. No continuity of α is imported or used. If $|t| < d(g, O)$, then $g \equiv 1$ on $O_{|t|}$, and (P) with $e^{-itH(g)}\Omega_g = \Omega_g$ gives, for $A, B \in \mathfrak{A}(O)$, the identity of Fig. 6:

$$\omega_g\big(A\alpha_t(B)\big) = \langle \Omega_g, Ae^{itH(g)}Be^{-itH(g)}\Omega_g \rangle = \langle A^*\Omega_g, e^{itH(g)}B\Omega_g \rangle = G_g(t).$$

[R, proof of Proposition 12.1, p. 29]

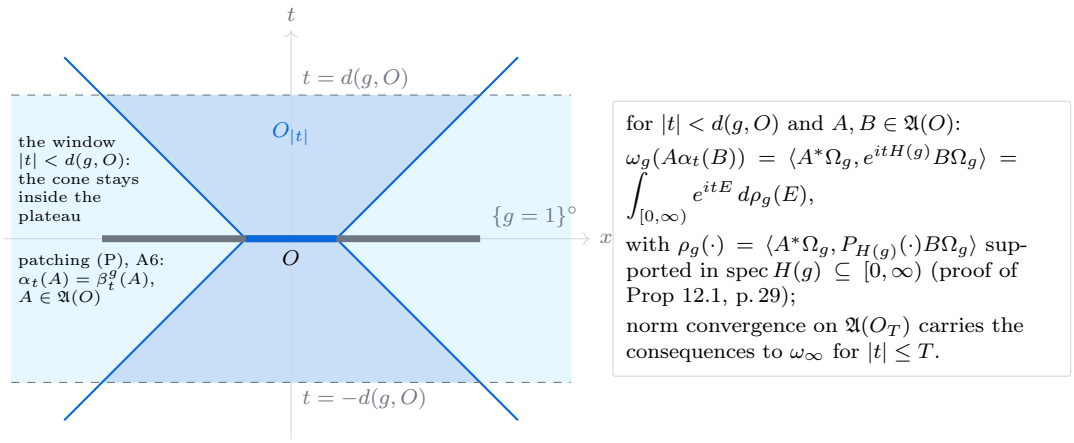


Figure 6: The identity window. The plateau interior $\{g = 1\}^\circ$ of a cutoff g on the axis $t = 0$, a bounded interval O inside it, the cone $O_{|t|}$, and the band $|t| < d(g, O)$ on which the patching identity (P) of [R, A6, p. 9] makes $\alpha_t(A) = \beta_t^g(A)$ for $A \in \mathfrak{A}(O)$. Inside the window the spectral condition is an identity at each cutoff; norm convergence on each $\mathfrak{A}(O_T)$ carries it to ω_∞ (Proposition 12.1). α is Glimm–Jaffe's patched dynamics, imported in A6(ii); the record does not construct it. Drawn from [R, A6, eq. (P), p. 9] and the proof of [R, Proposition 12.1, p. 29]; nothing is computed. Illustration, not evidence.

Here $G_g(t) := \int e^{itE}\, d\rho_g(E)$, where, with $P_{H(g)}$ the spectral measure of $H(g)$, $\rho_g(\cdot) := \langle A^*\Omega_g, P_{H(g)}(\cdot)B\Omega_g\rangle$ is a complex measure with $\operatorname{supp}\rho_g \subseteq \operatorname{spec} H(g) \subseteq [0, \infty)$: inside the window, each two-point function is the Fourier–Stieltjes transform of a measure on $[0, \infty)$. For $|t| \le T$, $A\alpha_t(B) \in \mathfrak{A}(O_T)$ has norm at most $\|A\|\|B\|$, so norm convergence on $\mathfrak{A}(O_T)$ makes these functions converge uniformly on $|t| \le T$, and the definition passes to the limit:

Proposition 12.1 of the record (p. 29)

If $(g_n) \subseteq \mathcal{G}$ satisfies $d(g_n, O) \to \infty$ for every bounded interval O, and $\omega_{g_n} \to \omega$ in norm on every $\mathfrak{A}(O)$, then ω is a ground state of $(\mathfrak{A}, \alpha)$ in the spectral sense: $\omega \circ \alpha_t = \omega$ for all t; the function $t \mapsto \omega(A\alpha_t(B))$ is continuous and bounded for all $A, B \in \mathfrak{A}$; and

$$\int_{\mathbb{R}} f(t)\, \omega(A\alpha_t(B))\, dt = 0$$

for every $f \in L^1(\mathbb{R})$ whose transform $\check{f}(E) = \int f(t)e^{itE}\, dt$ vanishes on $[0, \infty)$.

Translation invariance needs no averaging: $g \mapsto \tau_a g$ is an order isomorphism of $\mathcal{G}$ with cofinal range, so (T), $\omega_{\tau_a g} = \omega_g \circ \tau_{-a}$ [R, Lemma 2.8, p. 12], and Theorem 12.2(1) give $\omega_\infty \circ \tau_a = \omega_\infty$; parity follows likewise for even $\mathscr{P}$ [R, proof of Theorem 12.2, p. 30].

By Corollary 12.3, every w^* limit point of the averaged or unaveraged cutoff states of [3, §2] coincides with ω_∞ on each $\mathfrak{A}(O)$ [R, Corollary 12.3, p. 31]. Its proof takes GJ3's cutoff function g in $\mathcal{G}$ and applies Theorem 12.2(1), uniformly in the shift, to each shifted cutoff in the average. So ω_∞ is canonical for the cutoff family; its uniqueness is not claimed (§8.1). *In the record:* A6, p. 9; Lemma 2.8, p. 12; Theorem 11.1 with eqs. (31)–(32), Propositions 11.2 and 12.1, Theorem 12.2 and Corollary 12.3, pp. 28–31.

8 Boundaries, and Part II

Not unique, not sharp, not beyond weak coupling; and Part II is a different question.

8.1 What is not claimed

(a) Uniqueness. Theorem 12.2, conditional on U_{GJS}, identifies the limit of this cutoff family; it does not show that every locally normal ground state of $(\mathfrak{A}, \alpha)$, in the spectral sense of [R, Proposition 12.1, p. 29], equals ω_∞ [R, §1.3(a), p. 4]; [R, §14, *Uniqueness*, p. 33].

(b) Sharp rate. Ψ_{m_1} is super-polynomial but not exponential [R, §1.3(b), p. 4]. A proof of a sharp rate would require "an additional resolvent or Combes–Thomas input not supplied here" [R, §14, *Sharpness*, p. 33]. The record reports, from its working note `m2/` (a proof sketch with numerics, outside the document of record), that in the exactly solvable quadratic model the rate is $\mathrm{poly}(d)\, e^{-2md}$, with m that model's mass.

(c) Beyond weak coupling. No claim is made outside $0 \le \lambda/m_0^2 < \varepsilon_{\mathscr{P}}$ [R, §1.3(c), p. 4]. Theorem 13.2 is conditional on U_{GJS} and on Hypothesis 13.1 [R, Remark 13.3, p. 32]; [R, §14, *Beyond weak coupling*, p. 33], whose four items (B1)–(B4) are all load-bearing [R, §13, p. 31]. For even $\mathscr{P}$ at weak coupling, (B1)–(B4) cannot all hold; the proof uses the uniform gap through Lemma 9.3 and Theorem 12.2 (1), (3), (4), hence also (M_2) and U_{GJS}.

(d) Not addressed. A gap for a GNS Hamiltonian of ω_∞ is not addressed: item (5) concerns the cutoff Hamiltonians $H(g)$, $g \in \mathcal{C}_{\mathrm{H}}$. Nor is ω_∞ identified with the Euclidean (Wightman) theory of GJS; no reverse Osterwalder–Schrader reconstruction is used [R, abstract, p. 2]. Lorentz covariance is not discussed.

8.2 Part II: open at gate (II.11)

Nothing in Part I depends on Part II; §14 of the record states that the two problems have "no mathematical dependence in either direction" [R, §14, p. 33]. Part II asks a different question: uniformity in a lattice or wavelet resolution, in the operator-algebraic renormalisation programme over the wavelet maps of [16] [R, §1.3(d), p. 4], [R, §14, p. 33]. Its gate (II.11) asks that the lattice ground states, evaluated on each fixed coarse Weyl generator, converge as the resolution is refined to the vector state of the ground vector of a continuum torus operator constructed in phase V3 (`part_ii/gate_II11_scoping.pdf`, §1); the scoping pass "proves no part of (II.11)". Already for the free model, no finest-scale local-state-norm error of order $\varepsilon_N^{\vartheta}$, $\vartheta > 0$, is possible [R, §14, p. 33], so the comparison is made at a fixed coarse scale. Status per claim, verbatim from `docs/status.md`:

claim	status
finest-scale diagonal norm convergence	**disproved**
exact finite-torus interacting model and ground state	**constructed**
interacting fixed-coarse projective cluster state along a cofinal subnet	**proved unconditionally**
MMST free fixed-coarse / projective convergence	proved in the source, **strengthened here** to an explicit D4 power rate
exact free gap, group-velocity bound, Wick logarithm	**proved** in the Lamport note
Battle–Federbush as a source for OAR-uniform CE1/CE2	**not available** — the cited note treats different phase-cell variables
full-sequence interacting convergence and identification with the continuum $P(\varphi)_2$ state	**open**, exactly at the coarse Weyl characteristic-functional limit
interacting scale-uniform gap, local second moment, Lieb–Robinson	**open**

claim	status
the cited Nachtergaele–Raz–Schlein–Sims theorem	**inapplicable** to the Wick quartic: it assumes $V' \in L^1(\mathbb{R})$
the former path $(1-s)H_N + sH$	**undefined** without a common Hilbert-space/form realization; removed from the corrected program

The record is not consistent about this gate: `strategy.md` and the ledger of phase V4 record (II.11) as closed at the stated import scope, `part_ii/README.md` says that phase closure was not yet reached, and the repository README says the gate is open; this companion uses the README's wording.
In the record: abstract, p. 2; §1.3, p. 4; §13 with Hypothesis 13.1, Theorem 13.2 and Remark 13.3, pp. 31–32; §14, p. 33.

9 The record and how to check it

Every import is pinned to a page of its source; nothing here has been refereed.

The record's repository is "an experiment in open science", and the work in it is AI-assisted: the author drafted it with Anthropic's Claude as an interactive assistant and published the working method, including the rounds that found errors in earlier rounds' repairs (`README.md`, "What this is"). Human verification is ongoing.

Gates. The gate table of `docs/status.md` lists the Part I gates, all passed on 10 August 2026:

- G0: it compiles; citations and references resolve; every import names its users;
- G1/G2: each step of the two structured proof reconstructions maps to one environment or display; no unmapped step, no orphan;
- G3a/G3b: constants recomputed by hand; a forbidden-string check;
- G4: citations and dossier entries correspond one to one; regression greps;
- G5: three `pdflatex` passes, nothing unresolved, every page inspected;
- G6: an adversarial pass; no confirmed finding, or every one fixed and re-checked.

The adversarial passes in `strategy.md`, which the README says the assistant ran, are published.

Corrections. The corrections history includes audit item P1 (a missing premise, now (B4) of Hypothesis 13.1), S1–S3 (citation pinpoints: FKN transfer, hypercontractivity, domain smoothing) and L1–L6 (local repairs, from a cell count to a spectral-parameter range) [R, Appendix B, p. 35], and gate R2 (the printed-norm prefactor, read from the page image of GJS p. 594) [R, Appendix B, p. 34]. The scope amendment of 5 September 2026 gave U_{GJS} (A8(vi)) the status described in §4; "No proof in the manuscript changed" [R, title page, p. 1].

C5. Audit item C5 (§4) is open [R, Appendix B, p. 35]: all nine verdicts of `part_i/C5_reading_record.md` are blank. It checks transcription, not mathematics; closing it does not change the status of U_{GJS} (`docs/status.md`).

How to check. Every transcribed display of `part_i/main.tex` carries a `% src:` comment naming its origin, and every import names its pinpoint source and the page images documenting it in the audit dossier [R, §2.4, p. 7]. The page images are third-party material and are not redistributed; `README.md` names a script that regenerates those of the 16 August 2026 audit from a reader's own copies of the sources. Statement and equation numbers here come from `part_i/main.aux`.
In the record: title page, p. 1; §2.4, p. 7; Appendix B, pp. 34–35.

Appendix A. Where every statement lives

One row per numbered statement of the record, in order, with its number, title and printed page. *Depends on*, read from the proofs: U, consumes U_{GJS} directly or through a statement of class U, including via (M_2) or (G), hence conditional on U_{GJS}; G_γ, assumes (G_γ), nothing of class U; P, uses a printed A8 item (CE2, the norms, or the dilation) and nothing of class U; –, none of these; import, A1–A8 and Source Facts 2.2–2.4. Last column: the section of this companion that treats the statement, or why it is routine or bookkeeping.

no.	title	page	depends on	treated in, or why routine
2.1	cutoff classes	6	–	§1
A1	cutoff construction	7	import	§§4, 5
A2	Feynman–Kac–Nelson transfer	8	import	§§4, 5
A3	finite-volume stability	8	import	§4
A4	free hypercontractivity	8	import	§4
A5	domain smoothing; common core	8	import	§4
A6	free propagation; patching; the dynamics α	9	import	§§4, 6, 7
A7	covariance data	9	import	§4
A8	the GJS package	9	import	§4
2.2	CE2 = [7, Thm. 1.1.7], p. 594	10	import	§§3, 4, 5
2.3	CE1 = [7, Thm. 1.1.8], p. 595	10	import	§4
2.4	[7, p. 629, §4]	10	import	§4
2.5	conventions lock	12	–	Wick order equals C-order
2.6	domain smoothing at the ground state	12	–	$\Omega_g \in D(V(h))$ in two lines
2.7	cutoff propagation, uniformly in g	12	–	§6
2.8	covariance and parity of the cutoff states	12	–	§7
3.1	class compatibility	13	–	§6
3.2	dimensionless weak-coupling reduction	13	P	dilation into the printed region
3.3	coordinate convention and transport of the conclusions	14	P	on working coordinates
4.1	slab FKN; bounded nonnegative arguments	15	–	§5
4.2	signed bounded extension	15	–	bounded, then signed
4.3	unbounded extension	15	–	unbounded, by truncation
4.4	slab vacuum; simplicity only	16	–	§5
5.1	uniform local interaction moment (M_2)	16	U	§5
5.2	φ^4 instance	17	U	φ^4 arithmetic
6.1	cell count	17	–	cell counting
6.2	GJS norms of polynomial observables	18	P	GJS norms
6.3	row independence	18	P	GJS norms
7.1	moment transfer	18	U	§5
7.2	one-point limit	18	U	§5
7.3	two-point lower semicontinuity	19	U	§5
7.4	CE2 across a horizontal strip	19	P	§5
7.5	semigroup decay of centered observables	20	U	§5
7.6	exponential moments	20	U	§5
7.7	density of polynomial vectors	21	U	§5
7.8	uniform gap (G)	21	U	§5
8.1	path regularity of the semigroup	22	–	Hölder bookkeeping
8.2	projection and eigenvector regularity; derivative formulas	23	G_γ	§6
9.1	exact inverse-energy filter	24	–	§6
9.2	spectral separation	25	–	§6
9.3	clustering with one unbounded local argument	25	G_γ	§6

no.	title	page	depends on	treated in, or why routine
10.1	state derivative	26	G_γ	§6
10.2	localization of the perturbation	26	U	§6
10.3	two-regime bound	27	U	§6
10.4	one-path estimate	27	U	§6
10.5	the rate function is super-polynomial	28	U	§6
11.1	full-net convergence with rate	28	U	§7
11.2	local normality	29	U	§7
12.1	exact spectral passage	29	–	§7
12.2	Main Theorem	30	U	§1
12.3	the Glimm–Jaffe vacuum is canonical at weak coupling	31	U	§7
13.1	structure of a broken phase	31	–	§8.1
13.2	conditional exclusion of a two-phase structure	31	U	§8.1
13.3	scope of the phase-transition literature	32	–	§8.1
C.1	consequence of the audited source mechanisms	36	–	§4
C.2	normalisation	37	–	§4
C.3	Wick lower bound; the $h \geq 0$ entry	37	–	not consumed by Thm 12.2
C.4	ultraviolet difference; the $h \leq 1$ entry	37	–	not consumed by Thm 12.2
C.5	h-uniform stability	38	–	§4
C.6	why the class is what it is	38	–	source-mechanism audit
C.7	coverage of every CE application	38	U	§4

Appendix B. Notation

Symbols as the record fixes them, with their defining places; Theorems 5.1, 7.8 and 11.1, Proposition 10.4 and Lemma 10.5, cited below, are conditional on U_{GJS}.

symbol	meaning
$\mathcal{H}$, m_0	Q-space over the time-zero free field of mass $m_0 > 0$ [R, §2.1, p. 6]
$\mathfrak{A}(O)$, $\mathfrak{A}$	von Neumann algebra of Weyl operators on O; norm-closed union [R, §2.1, p. 6]
$V(h)$, $\mathscr{P}$	$\lambda \int h(x) :\mathscr{P}(\varphi_0(x)):_C dx$; $\mathscr{P}(\xi) = \sum_{n=0}^{2p'} a_n \xi^n$ [R, §2.1, p. 6]
$K(g)$, $E(g)$, $H(g)$	$H_0 + V(g)$; $\inf \operatorname{spec} K(g)$; $K(g) - E(g)$ [R, §2.1, p. 6]
Ω_g, ω_g	normalised ground vector of $H(g)$; its vector state [R, §2.1, p. 6]
β_t^g, α	$\operatorname{Ad} e^{itH(g)}$ [R, §2.1, p. 6]; patched dynamics with (P) [R, A6, p. 9]
$\mathcal{C}_{\mathrm{E}}$, $\mathcal{C}_{\mathrm{H}}$, $\mathcal{G}$	cutoff classes; $\mathcal{G}$ directed by inclusion of plateau interiors [R, Definition 2.1, p. 6]
O_r, $d(g, O)$	$\{x : \operatorname{dist}(x, O) < r\}$; $\operatorname{dist}(O, \mathbb{R} \setminus \{g = 1\}^\circ)$ [R, Definition 2.1, p. 6]
$h_{T,g}$, dq_h	slab cutoff $\mathbf{1}_{[-T,T]} \otimes g$ [R, A2, p. 8]; interacting Euclidean measure [R, A3, p. 8]
$\|\cdot\|_{\mathrm{GJS}}$	printed observable norm (1.1.7)–(1.1.9) of [7] [R, A8(iii), p. 10]
m_1	the GJS decay constant (A8(iv)); also the gap of Thm 7.8 [R, A8(iv), p. 10]
$\varepsilon_{\mathscr{P}}$	λ_*/M_*^2; the region is $0 \leq \lambda/m_0^2 < \varepsilon_{\mathscr{P}}$ [R, §2.6, p. 13]
M	moment constant of (M_2) [R, Theorem 5.1, p. 16]
Ψ_γ, d_0	rate function, eq. (29) [R, Lemma 10.5, p. 28]; $d_0 = 4 + 2/\gamma$ [R, Proposition 10.4, p. 27]
ω_∞	the limit state [R, Theorem 11.1, p. 28]

References

[R] A. Stottmeister, *Removal of the spatial cutoff for weakly coupled $P(\varphi)_2$: norm convergence of the ground-state net on the quasi-local algebra. Part I — unified manuscript*, document of record `part_i/main.pdf`, 40 pp., version of 5 September 2026, https://github.com/alexander-stottmeister/pphi2-cutoff-removal.

[1] J. Glimm, A. Jaffe, *A $\lambda\phi^4$ quantum field theory without cutoffs. I*, Phys. Rev. **176** (1968) 1945–1951.

[2] J. Glimm, A. Jaffe, *The $\lambda(\varphi^4)_2$ quantum field theory without cutoffs. II. The field operators and the approximate vacuum*, Ann. of Math. **91** (1970) 362–401.

[3] J. Glimm, A. Jaffe, *The $\lambda(\varphi^4)_2$ quantum field theory without cutoffs. III. The physical vacuum*, Acta Math. **125** (1970) 203–267.

[4] J. Glimm, A. Jaffe, *The $\lambda(\varphi^4)_2$ quantum field theory without cutoffs. IV. Perturbations of the Hamiltonian*, J. Math. Phys. **13** (1972) 1568–1584.

[5] J. Glimm, A. Jaffe, *Positivity and self-adjointness of the $P(\phi)_2$ Hamiltonian*, Comm. Math. Phys. **22** (1971) 253–258.

[6] J. Glimm, A. Jaffe, *Quantum Field Theory and Statistical Mechanics: Expositions*, Birkhäuser, Boston, 1985.

[7] J. Glimm, A. Jaffe, T. Spencer, *The Wightman axioms and particle structure in the $\mathscr{P}(\phi)_2$ quantum field model*, Ann. of Math. **100** (1974) 585–632.

[8] F. Guerra, L. Rosen, B. Simon, *The $P(\phi)_2$ Euclidean quantum field theory as classical statistical mechanics*, Ann. of Math. **101** (1975) 111–189, 191–259.

[9] L. Rosen, *A $\lambda\phi^{2n}$ field theory without cutoffs*, Comm. Math. Phys. **16** (1970) 157–183.

[10] B. Simon, *The $P(\Phi)_2$ Euclidean (Quantum) Field Theory*, Princeton University Press, Princeton, 1974.

[11] S. Albeverio, R. Høegh-Krohn, B. Zegarlinski, *Uniqueness and global Markov property for Euclidean fields: the case of general polynomial interactions*, Comm. Math. Phys. **123** (1989) 377–424.

[12] R. Bauerschmidt, B. Dagallier, H. Weber, *Holley–Stroock uniqueness method for the φ^4_2 dynamics*, arXiv:2504.08606 (2025).

[13] M. Rodriguez Zarate, T. Thiemann, *Hamiltonian renormalisation VIII. $P(\Phi)_2$ quantum field theory*, arXiv:2505.13030 (2025).

[14] M. B. Hastings, *Lieb–Schultz–Mattis in higher dimensions*, Phys. Rev. B **69** (2004) 104431.

[15] S. Bachmann, S. Michalakis, B. Nachtergaele, R. Sims, *Automorphic equivalence within gapped phases of quantum lattice systems*, Comm. Math. Phys. **309** (2012) 835–871.

[16] V. Morinelli, G. Morsella, A. Stottmeister, Y. Tanimoto, *Scaling limits of lattice quantum fields by wavelets*, Comm. Math. Phys. **387** (2021) 1299–1367.

Files of the record cited by path (all in the repository above): `README.md`, `docs/status.md`, `docs/imports.md`, `part_i/C5_reading_record.md`, `part_ii/README.md`, `part_ii/gate_II11_scoping.pdf`, `strategy.md`, `m2/m2.tex`, `docs/figures/make_figures.py`.